

Engineering Ethics

“Engineers, the Environment, and the Unknown”

Hochschule Darmstadt University of Applied Sciences

May 23, 2022

Agenda for today

1. Introduction
2. Risk Society
3. The Unknown – some examples
4. Debating about

Introduction

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- Final Exam
 - June 20th – 12:30 – 14:00h
- Saturday Seminar
 - June 11th – 09:30 – 15:30h
 - Room: 008 (Ground Floor) in Building A12 close to the REWE

Built to take your breath - Audi A8

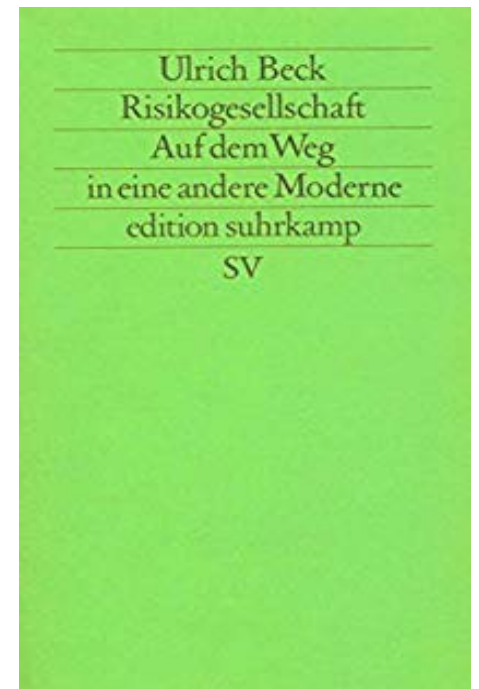


Risk Society

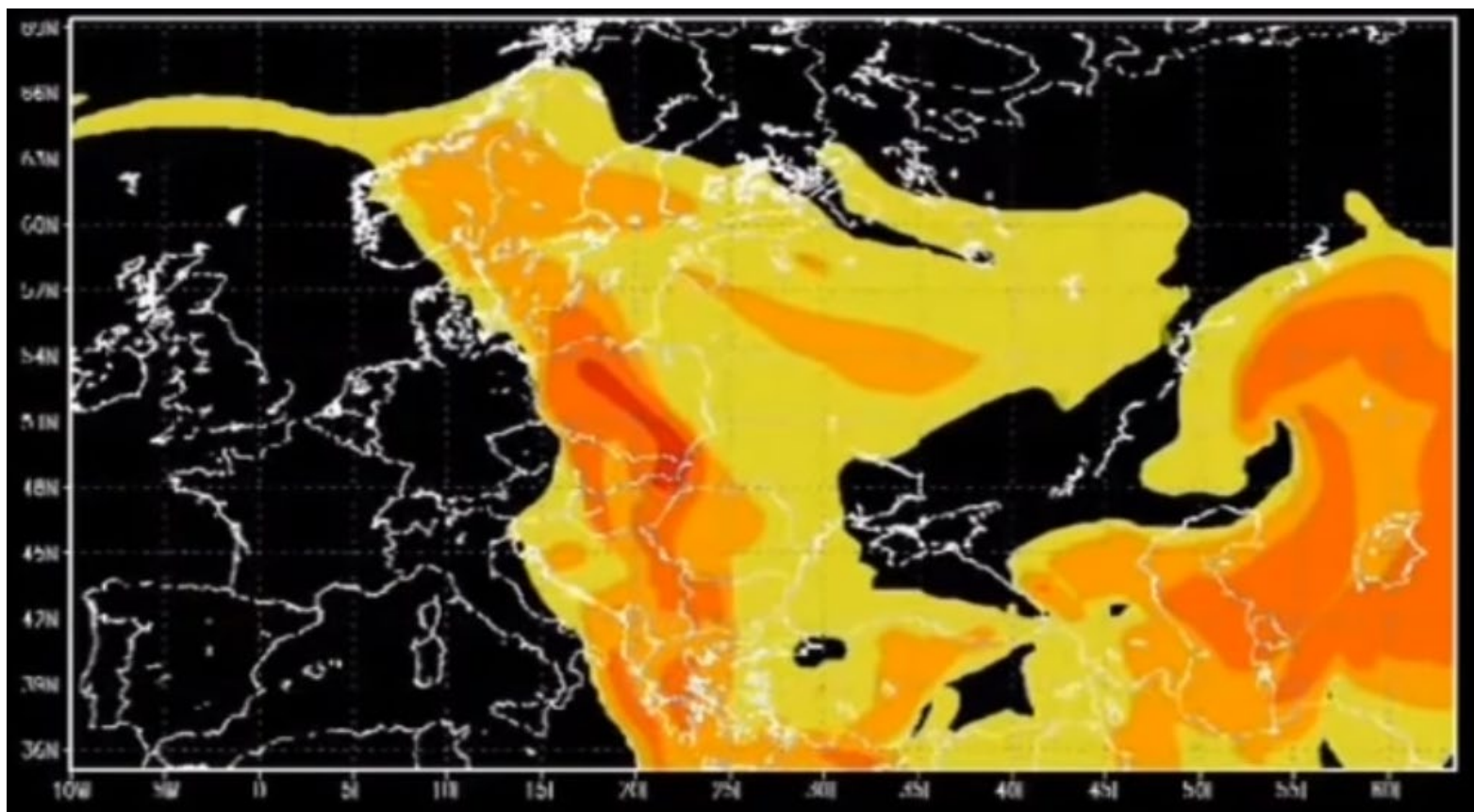


Ulrich Beck
(1944 – 2015)

"Much of what was still fought for argumentatively in writing - the imperceptibility of the dangers, their dependence on knowledge, their supranationality, the "ecological expropriation", the transformation of normality into absurdity, etc. - reads after Chernobyl like a flat description of the present." (1986: S.10)



Risk Society

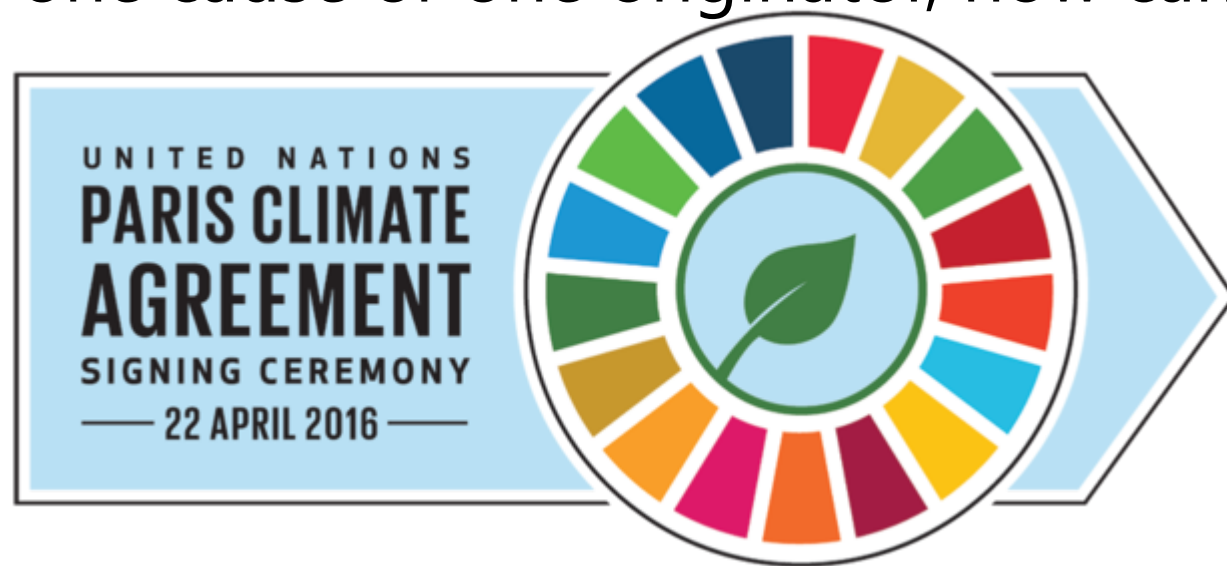


https://www.youtube.com/watch?v=mqu_l29WioM&list=RDmqu_l29WioM&start_radio=1&t=106

Risk Society

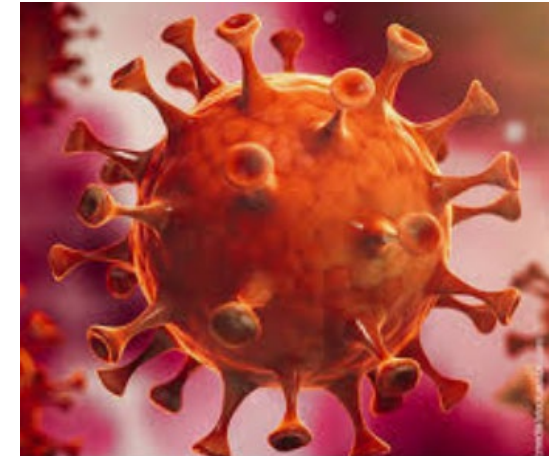
Central question for Beck

In view of the fact that the threats are increasingly global and cannot be traced back to one cause or one originator, how can a nation-state policy / society cope with these threats?



Risk Society

- SARS
- Bird flu
- Ozone depletion
- Harmful substances
- Global warming
- Climate change
- Covid-19
- Migration
- Chernobyl
- Ebola
- BSE
- Nuclear radiation
- Sweetwater scarcity
- Resource scarcity
- Rising sea levels
- Methane
- Cigarette smoke
-



Risk Society

Central question for Beck

The question of problem-solving ability becomes more acute insofar as ... the problems to be solved are not the result of a fundamental failure, but a side-effect of outstanding success.

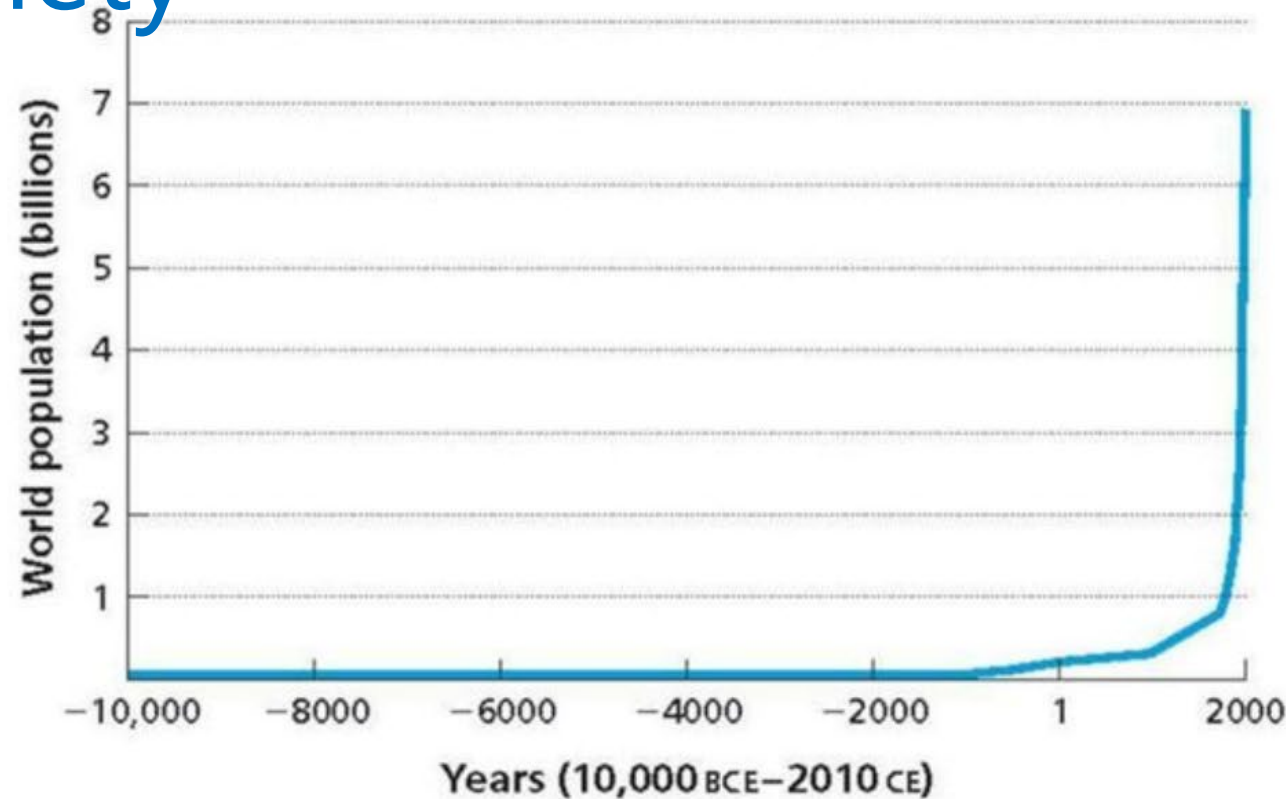


Photo:
Margaret Bourke-White
1937

"Your misery is the waste product of people without misery."

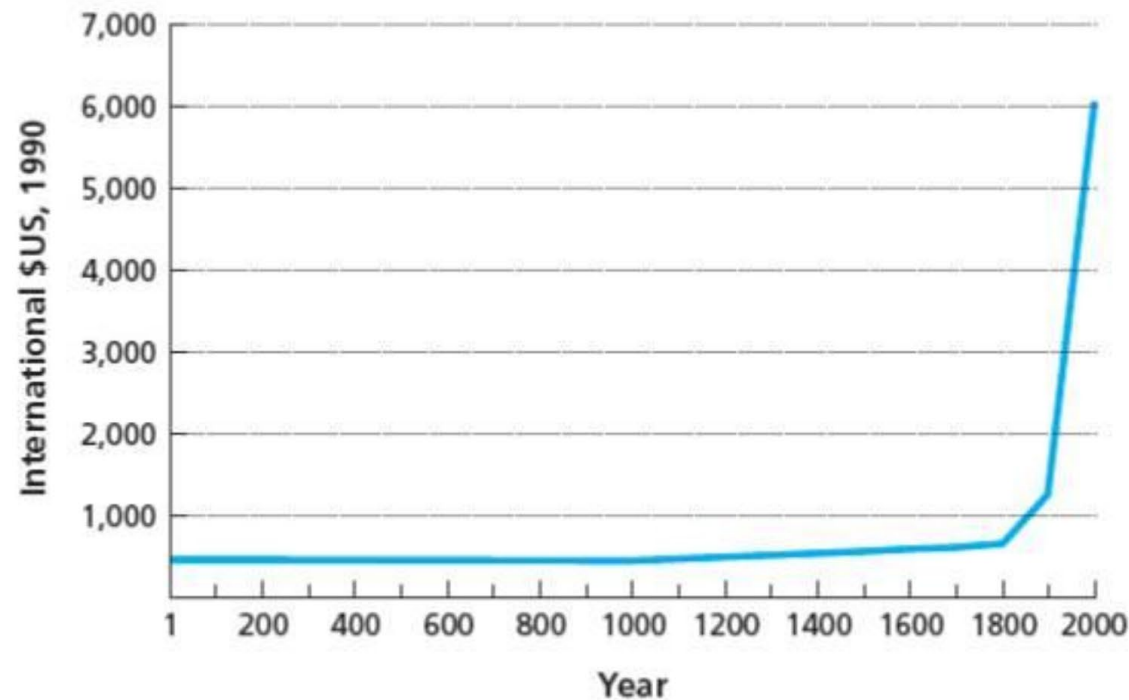
(John Steinbeck 1937:
"The grapes of Wrath")

Risk Society



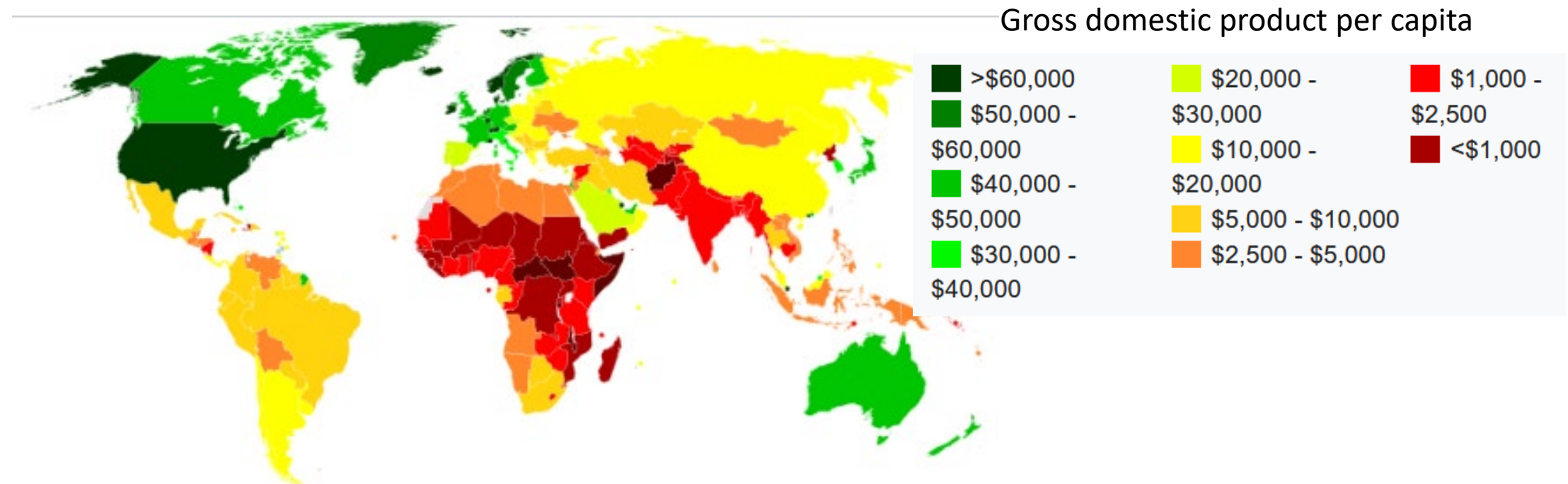
Source: Bolt, J., and J. L. van Zanden. 2013. "The First Update of the Maddison Project: Re-Estimating Growth Before 1820." Maddison Project Working Paper 4.

Gross World Product per Capita (1990 International Dollars)
Based on Maddison, The World Economy, 2001



Source: Bolt, J., and J. L. van Zanden. 2013. "The First Update of the Maddison Project: Re-Estimating Growth Before 1820." Maddison Project Working Paper 4.

Risk Society



The Unknown

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The Unknown



The Unknown - Products

- A well and helpful documentary

<https://www.ewg.org/skindeep/>

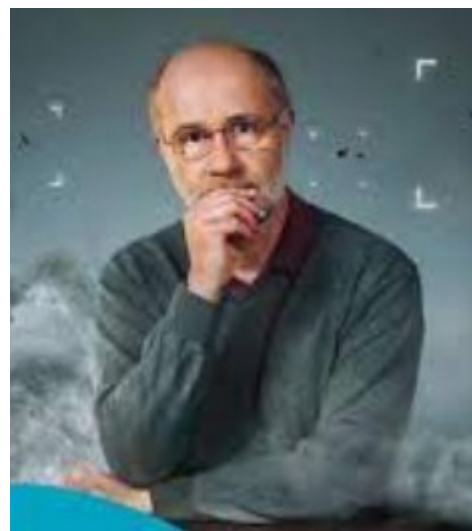
The Unknown - Technologies

- The society as a lab
 - The function of the lab
 - Containment
 - Scaling up
 - Hide away
 - If technologies are big, they need the society as a testbed

Debating about: **Can the economy (everything) grow without ruining the climate?**



Sarna Röser, born 1987.
Chairwoman of the young
entrepreneurs, Germany.
Representing young industry
leaders from family-owned
businesses.

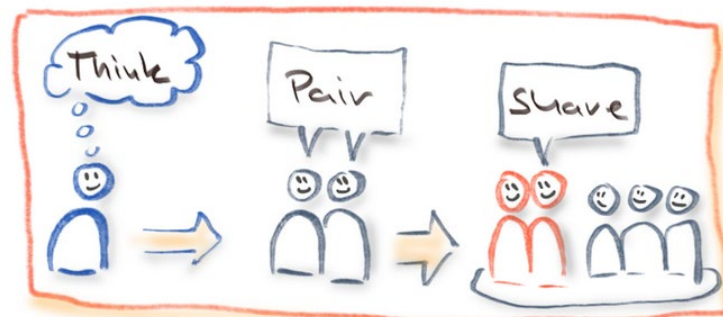


Harald Lesch, born 1960.
Astrophysicist and technology
writer

Debating about: **Can the economy (everything) grow without ruining the climate?**

- What do you think about the statement?

**„What we need are ethical
innovations, not technical ones.“**



Debating about: **Can the economy (everything) grow without ruining the climate?**

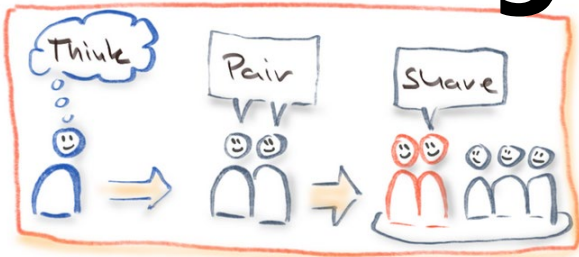
- Technical solutions and innovations or sacrifice?

“Christian social ethics places the individual at the center. Interference in the freedom of the individual, especially by the state, should be avoided as far as possible. In the end, it cannot be a question of making social policy through prohibitions! I would create the incentive that every person has an interest in helping and supporting.”

Debating about: Can the economy (everything) grow without ruining the climate?

- What do you think about the statement:

**„There is no example where (German)
industry has even approached truly
ecological goals on the basis of pure
self-responsibility.“**



Debating about: **Can the economy (everything) grow without ruining the climate?**

- What do you think about the statement:

„But we have to take all the people with us. A policy of bans and barricades that divides society does not help climate protection. We need market-based instruments. If we impose bans and restrict companies, we destroy their business models and jobs disappear.“